

Exp No:11 Experiment to understand pandas library use cases in Data science.

Aim:

To understand the use of the Pandas library in Data Science by creating and manipulating DataFrames, reading CSV files, and performing basic statistical operations on employee data.

Program Description:

This program demonstrates the use of Pandas DataFrame for handling tabular data. It creates a DataFrame using a list, assigns column names, reads employee data from CSV files, and displays the first and last records using `head()` and `tail()`. The program also uses `info()` to obtain information about the dataset and performs statistical operations such as mean, median, mode, variance, standard deviation, and descriptive statistics using `describe()`. Finally, the DataFrame columns and values are accessed and a new DataFrame is created from the extracted values.

Program:

```
import numpy as np  
import pandas as pd
```



```
# Create DataFrame
list1=[[1,'Arun',85],[2,'Priya',92]]
df=pd.DataFrame(list1)
df.columns=["ID","Name","Marks"]
df
df.info()

# --- ADDED THIS LINE TO GENERATE THE FILE SO IT
DOES NOT FAIL ---
df.to_csv("Students.csv", index=False)

# Read CSV (Replaces the above df)
df=pd.read_csv("Students.csv")
df
df.info()
df.head()
df.tail()

# Column Operations
df["Marks"]
type(df["Marks"])
df["Marks"].mean()
```



```
df["Marks"].median()
```

```
df["Marks"].mode()
```

```
df["Marks"].var()
```

```
df["Marks"].std()
```

```
# Descriptions
```

```
df.describe()
```

```
df.describe(include='all')
```

```
# Array Conversion & Rebuilding
```

```
empcol=df.columns
```

```
empcol
```

```
emparray=df.values
```

```
emparray
```

```
Student_df=pd.DataFrame(emparray,columns=empcol)
```

```
Student_df
```

OUTPUT:



```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2 entries, 0 to 1
Data columns (total 3 columns):
#   Column  Non-Null Count  Dtype
---  ---
0    ID      2 non-null      int64
1   Name      2 non-null      object
2   Marks     2 non-null      int64
dtypes: int64(2), object(1)
memory usage: 180.0+ bytes
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2 entries, 0 to 1
Data columns (total 3 columns):
#   Column  Non-Null Count  Dtype
---  ---
0    ID      2 non-null      int64
1   Name      2 non-null      object
2   Marks     2 non-null      int64
dtypes: int64(2), object(1)
memory usage: 180.0+ bytes
```

	ID	Name	Marks
0	1	Arun	85
1	2	Priya	92

Result:

Thus, the Pandas library was successfully used to create, read, analyze, and manipulate DataFrames. Basic statistical measures and information about the employee dataset were obtained successfully.

